The basics: Binary numbers

- Bases we will use
  - Binary: Base 2
  - Octal: Base 8
  - Decimal: Base 10
  - Hexadecimal: Base 16
- Positional number system
  - $101_2 = 1\times2^2 + 0\times2^1 + 1\times2^0$
  - $63_8 = 6\times8^1 + 3\times8^0$
  - $A1_{16} = 10\times16^1 + 1\times16^0$
- Addition and subtraction

$$\begin{array}{r} 1011 \\ +\ 1010 \\ \hline 10101 \end{array} \qquad \begin{array}{r} 1011 \\ -\ 0110 \\ \hline 0101 \end{array}$$

Converting decimal numbers to binary

- $53 = 32 + 16 + 4 + 1$
  $= 2^5 + 2^4 + 2^2 + 2^0$
  $= 1*2^5 + 1*2^4 + 0*2^3 + 1*2^2 + 0*2^1 + 1*2^0$
  = 110101 in binary
  = 00110101 as a full byte in binary

- $211 = 128 + 64 + 16 + 2 + 1$
  $= 2^7 + 2^6 + 2^4 + 2^1 + 2^0$
  $= 1*2^7 + 1*2^6 + 0*2^5 + 1*2^4 + 0*2^3 + 0*2^2 + 1*2^1 + 1*2^0$
  = 11010011 in binary

Converting binary numbers to decimal

- What is 10011010 in decimal?

$$\begin{aligned} 10011010 &= 1*2^7 + 0*2^6 + 0*2^5 + 1*2^4 + 1*2^3 + \\ &\quad 0*2^2 + 1*2^1 + 0*2^0 \\ &= 2^7 + 2^4 + 2^3 + 2^1 \\ &= 128 + 16 + 8 + 2 \\ &= 154 \end{aligned}$$

- What is 00101001 in decimal?

$$\begin{aligned} 00101001 &= 0*2^7 + 0*2^6 + 1*2^5 + 0*2^4 + 1*2^3 + \\ &\quad 0*2^2 + 0*2^1 + 1*2^0 \\ &= 2^5 + 2^3 + 2^0 \\ &= 32 + 8 + 1 \\ &= 41 \end{aligned}$$

- How do we write negative binary numbers?
- Historically: 3 approaches
  - Sign-and-magnitude
  - Ones-complement
  - Twos-complement
- For all 3, the most-significant bit (MSB) is the sign digit
  - 0 ≡ positive
  - 1 ≡ negative
- twos-complement is the important one
  - Simplifies arithmetic
  - Used almost universally

Binary → hex/decimal/octal conversion

- Conversion from binary to octal/hex
  - Binary: 10011110001
  - Octal: 10 | 011 | 110 | 001=2361_8
  - Hex: 100 | 1111 | 0001=$4F1_{16}$
- Conversion from binary to decimal
  - $101_2 = 1\times2^2 + 0\times2^1 + 1\times2^0 = 5_{10}$
  - $63.4_8 = 6\times8^1 + 3\times8^0 + 4\times8^{-1} = 51.5_{10}$
  - $A1_{16} = 10\times16^1 + 1\times16^0 = 161_{10}$

Decimal→ binary/octal/hex conversion

Binary

	Quotient	Remainder
56÷2=	28	0
28÷2=	14	0
14÷2=	7	0
7÷2=	3	1
3÷2=	1	1
1÷2=	0	1

Octal

	Quotient	Remainder
56÷8=	7	0
7÷8=	0	7

$56_{10}=111000_2$

$56_{10}=70_8$

- Why does this work?
  - $N=56_{10}=111000_2$
  - $Q=N/2=56/2=111000/2=11100$ remainder 0
- Each successive divide liberates an LSB (least significant bit)

Sign-and-magnitude

- The most-significant bit (MSB) is the sign digit
  - 0 ≡ positive
  - 1 ≡ negative
- The remaining bits are the number's magnitude
- Problem 1: Two representations for zero
  - 0 = 0000 and also –0 = 1000
- Problem 2: Arithmetic is cumbersome

Add		Subtract			Compare and subtract		
4	0100	4	0100	0100	– 4	1100	1100
+ 3	+ 0011	– 3	+ 1011	– 0011	+ 3	+ 0011	– 0011
= 7	= 0111	= 1	≠ 1111	= 0001	– 1	≠ 1111	= 1001

Ones-complement

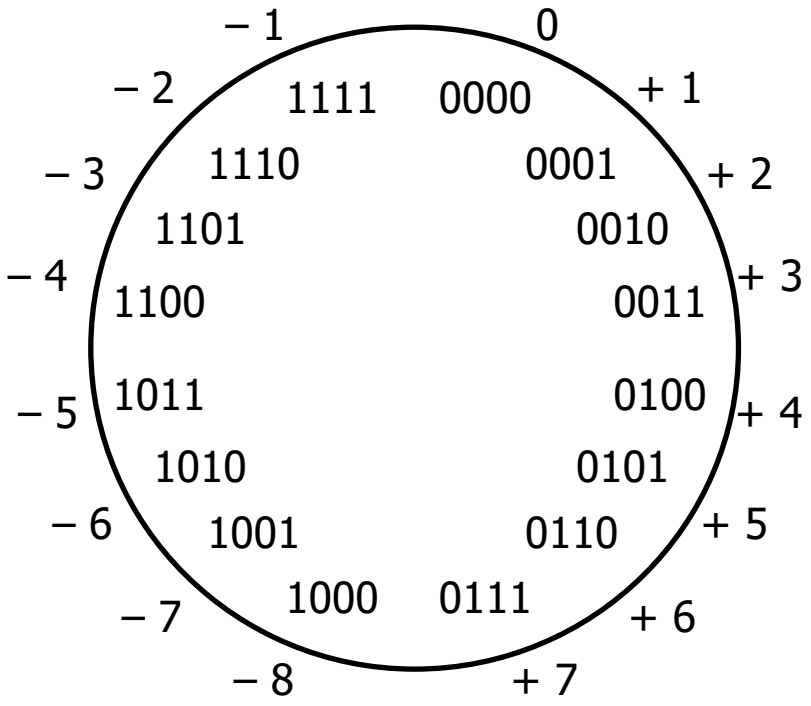
- Negative number: Bitwise complement positive number
  - $0011 \equiv 3_{10}$
  - $1100 \equiv -3_{10}$
- Solves the arithmetic problem

Add		Invert, add, add carry		Invert and add	
4	0100	4	0100	− 4	1011
+ 3	+ 0011	− 3	+ 1100	+ 3	+ 0011
= 7	= 0111	= 1	1 0000	− 1	1110
		add carry:	+1		
			= 0001		

- Remaining problem: Two representations for zero
  - 0 = 0000 and also –0 = 1111

Twos-complement

- Negative number: Bitwise complement plus one
  - $0011 \equiv 3_{10}$
  - $1101 \equiv -3_{10}$
- Number wheel



- Only one zero!
- MSB is the sign digit
  - 0 ≡ positive
  - 1 ≡ negative

Twos-complement (con't)

- Complementing a complement → the original number
- Arithmetic is easy
  - Subtraction = negation and addition
    - Easy to implement in hardware

Add		Invert and add		Invert and add	
4	0100	4	0100	– 4	1100
+ 3	+ 0011	– 3	+ 1101	+ 3	+ 0011
= 7	= 0111	= 1	1 0001	– 1	1111
		drop carry	= 0001		

Gray and BCD codes

Decimal Symbols	Gray Code
0	0000
1	0001
2	0011
3	0010
4	0110
5	0111
6	0101
7	0100
8	1100
9	1101

Decimal Symbols	BCD Code
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001

Review of Boolean algebra

- Not is a horizontal bar above the number
  - $\bar{0} = 1$
  - $\bar{1} = 0$
- Or is a plus
  - 0+0 = 0
  - 0+1 = 1
  - 1+0 = 1
  - 1+1 = 1
- And is multiplication
  - 0*0 = 0
  - 0*1 = 0
  - 1*0 = 0
  - 1*1 = 1

Basic logic gates

- Not
- And
- Or
- Nand
- Nor
- Xor